

DRY-RUN GUIDE | 03_minimum_size_subarray_sum.py

Source: Python\dsa_solutions\03_minimum_size_subarray_sum.py

CORE BEHAVIOR

Expands a positive-number window until target is met, then shrinks it to find the shortest qualifying range.

REPRESENTATIVE INPUT

Target 7; '[2,3,1,2,4,3]'.

STEP TRACE

A window reaches 8; after sliding/shrinking, '[4,3]' sums to 7 with length 2.

EXPECTED OUTPUT / RETURN

Minimum length '2'.

ACTUAL SOURCE EXCERPT

Verbatim excerpt from Python\dsa_solutions\03_minimum_size_subarray_sum.py; line numbers refer to the original file. No source was modified.

```
001 | import sys
002 |
003 |
004 | def main():
005 |     data = list(map(int, sys.stdin.read().split()))
006 |     if not data:
007 |         return
008 |     count = data[0]
009 |     numbers = data[1:1 + count]
010 |     target = data[1 + count]
011 |     left = 0
012 |     current_sum = 0
013 |     best_length = count + 1
014 |     for right, number in enumerate(numbers):
015 |         current_sum += number
016 |         while current_sum >= target:
017 |             best_length = min(best_length, right - left + 1)
018 |             current_sum -= numbers[left]
019 |             left += 1
020 |     print(0 if best_length == count + 1 else best_length)
021 |
022 |
023 | if __name__ == "__main__":
024 |     main()
025 |
```

DRY-RUN NOTE

Values are representative. For function-only modules the guide uses an illustrative direct call and explains its return; quirks/errors describe actual behavior.